

Hossein Molavi

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EXPERIENCE

SCADABLE

Technical Co-Founder

Waterloo, ON

Sept 2026 – Present

- Building AI-native compliance software that examines customer codebases, databases, infrastructure, and operations to identify gaps, remediate issues, and preserve audit evidence.
- Built the Harness, SCADABLE's agentic decision layer, to run repository extraction and multi-step model workflows, monitor execution, and record auditable verdicts while keeping non-judgment services deterministic.
- Helped close and onboard two of SCADABLE's three clients through four customer meetings, explaining the product and technical vision and surfacing compliance pain points; worked with CPA audit firms and legal counsel on contract terms.

onsemi

Wireless Firmware Developer

Waterloo, ON

Sept 2025 – Aug 2026

Contractor (Apr 2026 – Aug 2026)

- Resolved DMA and timing race conditions in the RSL20 NFMI receive path, improving hearing-aid audio integrity.
- Built a self-validating AI agent pipeline (CMake, J-Link) that moves a Jira ticket through code generation, concurrent multi-device build and flash, on-target log and unit-test validation, and PR submission; cut routine turnaround from 1–2 days to roughly 15 minutes.

Intern (Sept 2025 – Apr 2026)

- Built HFP/A2DP audio interfaces with mSBC/CVSD codecs and hardened NFMI ear-to-ear delivery with ACK-based retransmission.
- Wrote 580+ CppUTest tests for the NFMI application layer, achieving 100% code coverage with gcov.

Asymptote Labs Inc – Anti-Drone Defense Systems

Co-Founder & Firmware Lead

Waterloo, ON

Apr 2025 – Apr 2026

- Led a five-person engineering team building a real-time computer-vision turret integrating YOLO detection with precision motor control.
- Designed a four-byte UART protocol and FOV-calibrated control loop that converted detected pixel offsets into stepper-motor commands.

Wrmth Corp.

Firmware Developer

North Bay, ON

Jan 2025 – Apr 2025

- Built ESP32 firmware with secure BLE/Wi-Fi, MQTT/AWS IoT connectivity, and OTA updates within 512KB RAM; resolved critical race-condition memory leaks with JTAG.

EDUCATION

University of Waterloo

Bachelor of Applied Science in Computer Engineering (Honours, Co-op)

Waterloo, ON

Sept 2021 – Aug 2026

TECHNICAL SKILLS

Languages: C, C++, Python, Bash/Shell, Verilog, ARM Assembly

Systems: Agentic workflows, embedded systems, FreeRTOS, Zephyr, Linux, AWS IoT Core, Docker

Tools: Git, CMake/Make, GDB/JTAG/J-Link, CppUTest, gcov, Pytest, KiCad

PROJECTS

Real-Time Operating System | C, ARM Assembly, STM32

- Built a preemptive RTOS supporting 16 concurrent tasks with EDF scheduling, PendSV/SVC context switching, and a 32KB buddy allocator.

RideSafe – Bike Safety System | ESP32-S3, BLE (NimBLE), FreeRTOS, SPI/DMA

- Built a dual-ESP32 cyclist-awareness system with radar sensing, secure BLE, haptic alerts, and an 80MHz DMA-driven display pipeline.